

Explicit Behavioral Embedding Method for Generating Explanations of Insider Threat Events

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Abstract—Insider threats pose significant risks to organizational security; however, many detection models lack transparency, which hinders their practical deployment due to high false-positive rates. We propose a novel framework that integrates behavioral embeddings with SHAP (Shapley Additive Explanations) to enhance the explainability of unsupervised anomaly detection models. By modeling temporal and semantic aspects of user actions, our approach generates interpretable visualizations of anomalous behaviors. Applied to the CERT v4.2 Insider Threat Dataset, the framework effectively identifies and explains anomalies, such as unauthorized data transfers and malicious system activities, through clustering and feature-importance analysis. Our results demonstrate that explainable models help reduce analyst workload by filtering false positives and providing actionable insights. This approach enhances decision-making, facilitates compliance with transparency regulations such as GDPR, and addresses the trade-off between detection accuracy and interpretability, making it a valuable tool for real-world insider threat detection.

Index Terms—anomaly, malicious actions, insider threat, user behavior, explainability, autoencoder

I. INTRODUCTION

Insider threats, malicious or unintentional actions by authorized individuals, pose severe risks to organizational security due to their access to sensitive systems and data [1]. Unlike external attackers, insiders are harder to detect, as their legitimate access masks suspicious activities, leading to significant harm, such as data breaches or resource sabotage [2], [3]. Current machine learning-based detection systems, while effective at identifying behavioral anomalies, often produce opaque outputs, generating numerous false positives that burden security analysts [4], [5]. This lack of interpretability limits trust and practical deployment, especially in high-stakes environments requiring transparency, such as those governed by GDPR [6].

To address these challenges, we propose an explainable insider threat detection framework that combines nonlinear

behavioral embeddings with SHAP (Shapley Additive Explanations). Our approach models temporal and semantic user actions, uses unsupervised anomaly detection via dimensionality reduction and clustering, and generates interpretable explanations of detected anomalies. Tested on the CERT v4.2 Insider Threat Dataset, our method visualizes anomalous behaviors, such as unauthorized data transfers, enabling analysts to filter false positives efficiently and make informed decisions. This framework balances accuracy and interpretability, enhancing trust and compliance with regulatory requirements.

This paper is organized as follows: Section II reviews related work, Section III details the methodology, Section IV discusses nonlinear behavioral modeling, Section V presents the results and analysis, and Section VI provides a discussion and conclusion.

II. RELATED WORK

Insider threats, stemming from authorized individuals' malicious or unintentional actions, challenge organizational security due to their legitimate access to sensitive systems. Liu et al. [1] survey detection techniques, noting that supervised methods like Random Forests and Support Vector Machines excel with labeled datasets like CERT, but unsupervised methods, such as clustering, are vital for identifying novel threats when labeled data is scarce. Legg et al. [7] demonstrate unsupervised role-based profiling to detect behavioral anomalies without prior labeling, emphasizing the need for robust anomaly detection.

However, the opacity of machine learning models hinders their practical use, as security analysts struggle to interpret high false-positive rates [4]. Goodman and Flaxman [6] stress the importance of explainability in high-stakes domains, aligning with GDPR's transparency requirements. To address this,

Ribeiro et al. [8] propose LIME, a model-agnostic technique generating local explanations by approximating complex models with interpretable ones. Similarly, Lundberg and Lee [9] introduce SHAP (Shapley Additive Explanations), a unified framework that assigns feature importance and enhances model transparency. These methods have been adopted in insider threat detection to improve trust and decision-making.

Challenges persist, including noisy or incomplete data and adversarial evasion, as noted by Homoliak et al. [3]. Arrieta et al. [10] highlight the accuracy-interpretability trade-off, where complex models like deep neural networks sacrifice explainability for predictive power. Our work builds on these foundations, integrating unsupervised behavioral embeddings with SHAP to provide interpretable anomaly detection, addressing both transparency and practical deployment needs in insider threat scenarios.

III. METHODOLOGY

Our framework for explainable insider threat detection integrates three components: (a) nonlinear behavioral modeling to capture user actions, (b) anomaly detection using dimensionality reduction and clustering, and (c) SHAP-based explanations for interpretable outputs. This approach strikes a balance between detection accuracy and transparency, enabling security analysts to interpret and act on anomalies detected in the CERT v4.2 Insider Threat Dataset.

A. Nonlinear Behavioral Modeling: To model user behavior, we process time-series data of user actions from the CERT v4.2 dataset, segmented into daily windows [11]. Each day's actions are represented as a probability vector $x_i \in \mathbb{R}^d$, where d is the number of logged action types (e.g., file access, USB activity). Frequencies are normalized to form a probability distribution, capturing temporal and semantic patterns [12]. Unlike models optimized solely for predictive performance, our approach prioritizes explainability by generating embeddings that highlight anomalous behaviors for analyst review. This model acts as a funnel, surfacing relevant actions while reducing cognitive overload on analysts, thus supporting efficient false-positive filtering.

B. Anomaly Detection: Anomaly detection leverages unsupervised techniques to identify deviations in user behavior. We apply Isomap, a nonlinear dimensionality reduction method, to project high-dimensional action vectors into a 2D space, preserving behavioral patterns [13]. These embeddings are then clustered using DBSCAN, which groups similar behaviors and flags outliers as potential anomalies based on local density [14]. For example, days with unusual actions (e.g., unauthorized data transfers) form sparse clusters distinct from normal behavior. This pipeline effectively isolates anomalies (e.g., days 147, 152, 154 in Scenario 1) in the CERT v4.2 dataset, for further analysis.

C. SHAP Explanations: To generate interpretable explanations, we employ Kernel SHAP on a neural network surrogate model trained to approximate DBSCAN's anomaly labels. This enables us to apply SHAP in an unsupervised setting, where the surrogate captures the decision boundary of DBSCAN

clustering. SHAP assigns importance scores to input features (e.g., USB disconnect, HTTP) by evaluating their marginal contributions to anomaly predictions. These scores help analysts understand which actions contributed most to flagged anomalies, such as USB usage in data exfiltration scenarios.

While this approach enables post-hoc explainability, it introduces a layer of approximation. The quality of explanations depends on the surrogate's fidelity to DBSCAN. Future work may explore direct explainability for clustering or fully differentiable alternatives.

D. Integrated Framework: The framework combines these components as follows: user actions are modeled as probability vectors, reduced to low-dimensional embeddings via Isomap, clustered by DBSCAN to detect anomalies, and explained using SHAP values. Figure 1 illustrates this pipeline, showing a 2D Isomap projection with DBSCAN clusters (blue for normal, red for anomalous) and SHAP feature importance for a sample anomaly. This integrated approach ensures high detection accuracy while providing interpretable outputs for analysts, addressing the trade-off between performance and transparency.

IV. NONLINEAR BEHAVIORAL MODELING

This section corresponds to component (a) of our proposed framework: the nonlinear behavioral modeling module that captures temporal and semantic aspects of user actions.

To perform behavioral analysis on user actions, we have access to a time series of user actions generated by the CERT insider threat model [15]. This time series can be subdivided into a window of daily actions, which can then be used to build a behavioral model for the user. There is a trade-off between building the most accurate model and building the most explainable model [9] [10]. Our approach in this work is to mitigate this trade-off by developing an explainable behavioral model. By using an explainable behavioral model, we aim to enable the display of anomalous events to the Security Analyst [16]. The model, therefore, acts as a funnel, surfacing the relevant user actions for the analyst to further analyze. There is, thus, a trade-off where we might choose to sacrifice model accuracy to sharpen its explainability characteristics. The trick is to use a trade-off that generates manageable data for the analyst, while also highlighting all the anomalies detected in the user actions.

To build this model, we determine a frequency of user actions for a particular day and convert it into a probability distribution by normalizing the frequencies.

Let $\mathbf{x}_i \in \mathbb{R}^d$ represent the probability vector of frequencies observed in the day [12] for the user, where d is the number of action-types that the user performs and are logged by the monitoring system.

$$\mathbf{X} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \dots \quad \mathbf{x}_N] \quad (1)$$

We then use this matrix to compute a correlation matrix between the probability vectors of different days. This can be achieved by computing the inner product of the probability

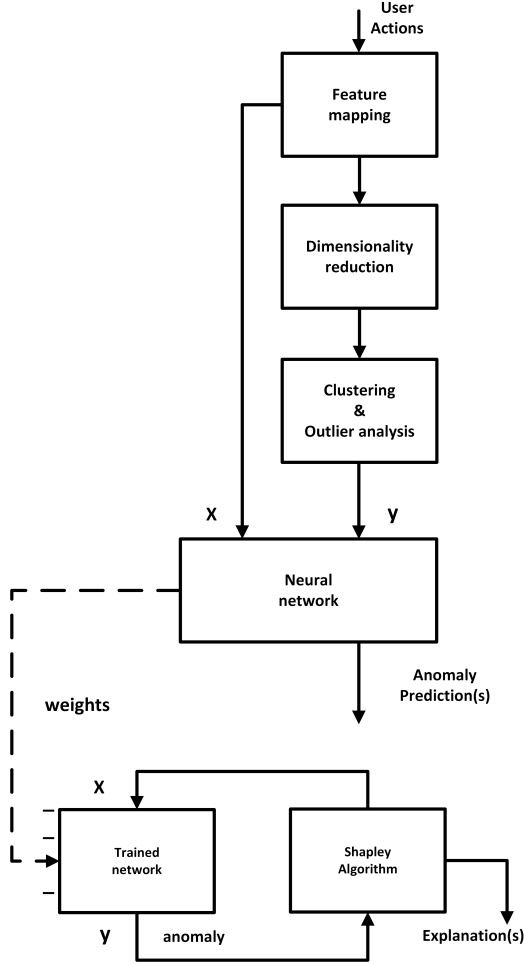


Fig. 1. Framework for Explainable Insider Threat Detection Using Nonlinear Behavioral Embeddings, Isomap-DBSCAN Anomaly Detection, and SHAP Explanations

vectors for each day with those of other days. We can accomplish this by performing an outer product operation on the data matrix.

$$\mathbf{C} = \mathbf{X}\mathbf{X}^\top \quad (2)$$

This matrix can be decomposed to identify a basis along which the probabilistic action vector can be projected, to create a mapping of different actions.

Let the dataset be $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$, where $\mathbf{x}_i \in \mathbb{R}^D$. The Euclidean distance between two points is:

$$d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\| \quad (3)$$

Construct the neighborhood graph by connecting points if they are within an ϵ radius:

$$\text{If } d_{ij} \leq \epsilon, \text{ then } \mathbf{x}_i \text{ and } \mathbf{x}_j \text{ are neighbors} \quad (4)$$

Compute the shortest path distances D_{ij} between all pairs of points:

$$D_{ij} = \text{Shortest path distance between } \mathbf{x}_i \text{ and } \mathbf{x}_j \quad (5)$$

Form the squared distance matrix:

$$D_{ij}^{(2)} = D_{ij}^2 \quad (6)$$

Compute the centering matrix:

$$\mathbf{H} = \mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top \quad (7)$$

Calculate the Gram matrix:

$$\mathbf{K} = -\frac{1}{2}\mathbf{H}\mathbf{D}^{(2)}\mathbf{H} \quad (8)$$

Perform eigenvalue decomposition:

$$\mathbf{K}\mathbf{v}_i = \lambda_i\mathbf{v}_i \quad (9)$$

Construct the low-dimensional embedding:

$$\mathbf{Y} = [\sqrt{\lambda_1}\mathbf{v}_1 \quad \sqrt{\lambda_2}\mathbf{v}_2 \quad \dots \quad \sqrt{\lambda_d}\mathbf{v}_d] \quad (10)$$

This allows for a decomposition that we can use to project user behaviors into the Cartesian map. Once projected, these behaviors can then be clustered using a clustering algorithm. The density of points within a cluster can be used as a proxy for distinguishing between anomalous and non-anomalous behavior.

The following steps, including dimensionality reduction via Isomap and clustering with DBSCAN, represent component (b) of our framework: the anomaly detection system.

This decomposition enables us to project user behaviors into a Cartesian map that preserves the intrinsic geometry of the behavior space. Our methodology closely follows the Isomap algorithm [13], a nonlinear dimensionality reduction technique that maintains geodesic distances on a manifold by approximating them through shortest paths on a neighborhood graph, as discussed in manifold learning theory [17]. This is essential for capturing complex temporal-semantic patterns in user behavior that may not be linearly separable in the original action space.

The choice of Isomap is informed by its proven effectiveness in uncovering latent structure in high-dimensional behavioral data, as highlighted in manifold learning literature [17]. Each user's daily action frequency vector is treated as a point in a high-dimensional space, and the sequence of transformations—from distance matrix computation to eigen decomposition—constructs a low-dimensional embedding that reveals underlying behavior modes.

These embeddings, once generated, serve as the input for downstream clustering algorithms to identify anomalies based on deviations from dense behavior clusters. This nonlinear projection is thus a crucial part of our framework, forming the foundation of the behavioral modeling component described in Section III.

V. RESULTS AND ANALYSIS

A. Scenario 1: Anomalous User # 103 ((BDV0168))

This CERT v4.2 database describes this scenario as follows: "A user who had not previously used removable drives or worked after hours begins logging in after hours, using a removable drive, and uploading data to wikileaks.org. The user leaves the organization shortly afterward". We will establish a specific anomalous user in this scenario, then apply our pipeline of events to the scenario, and examine the results. We begin by reviewing the Isomap-generated behavioral model for the user and clustering the points based on that behavior.

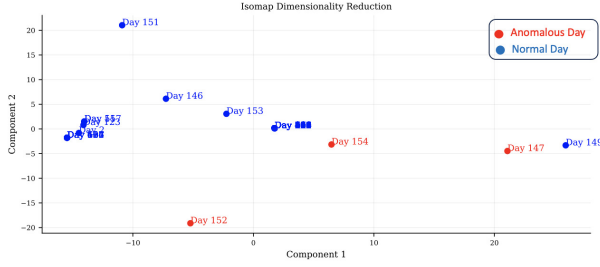


Fig. 2. Isomap Graph for (User 103)

Figure 2 shows a 2D representation using Isomap for dimensionality reduction. Each data point represents an observation from a particular day, labeled by its respective day number. The points are separated into two clusters:

Blue Cluster: The majority of the data points (days) fall in the blue cluster, which spans components 1 and 2. These points are primarily located in the upper left region, with a few scattered farther from the central cluster.

Red Cluster: A small set of days is highlighted in red (Days 152, 154, and 147), forming a separate cluster in the bottom-right and lower-left areas. These points are distinctly separate from the blue points, indicating that these days may represent anomalous or unique behavior compared to the majority.

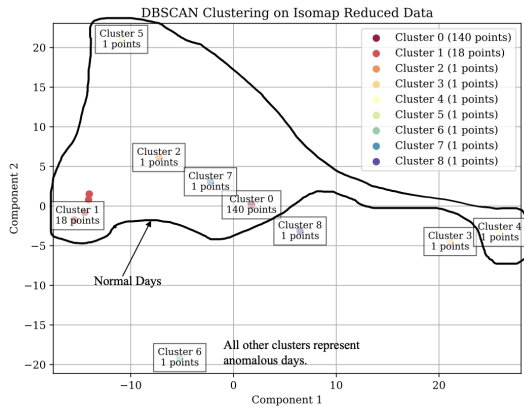


Fig. 3. DBSCAN Graph for (User 103)

Figure 3 shows the results of DBSCAN clustering on a dataset that has been reduced to two dimensions using Isomap. The visualization reveals a dominant Cluster 0 consisting of

140 points, which is the largest cluster. Cluster 1 contains 18 points and is the second-largest cluster in this instance. The remaining clusters, Clusters 2 through 8, each consist of 1 point, indicating that they may be noise points or isolated small clusters. This distribution suggests a high concentration of points in Cluster 0, with some sparse and scattered outliers.

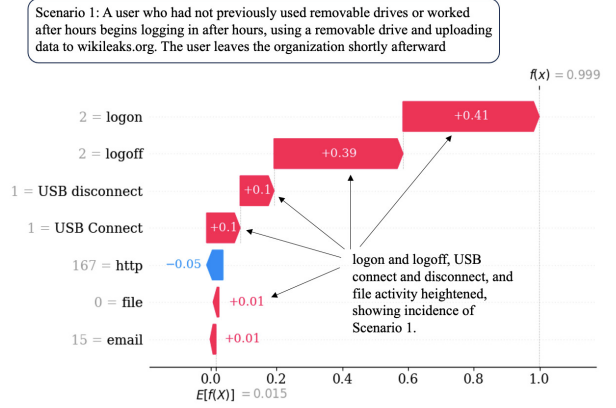


Fig. 4. SHAP Explanation for Scenario 1 (User 103) highlighting feature contributions.

Figure 4 shows SHAP values for day 154, revealing feature contributions to the model's prediction of $f(x) = 0.999$, deviating from the base value $E[f(x)] = 0.015$, indicating an anomaly.

Positive Contributions: Logon: +0.41, Logoff: +0.39, USB Connect: +0.1, USB Disconnect: +0.1, file: +0.01, email: +0.01

Negative Contributions: HTTP: -0.05

The final prediction score of the model is dominated by logon, logoff, USB Connect, and disconnect operations, as well as file and email actions, which collectively contribute to a high score. HTTP events marginally decrease the score.

B. Scenario 2: Anomalous User # 1 ((AAF0535))

This CERT v4.2 database describes this scenario as follows: "The user begins surfing job websites and soliciting employment from a competitor. Before leaving the company, they use a thumb drive (at markedly higher rates than their previous activity) to steal data".

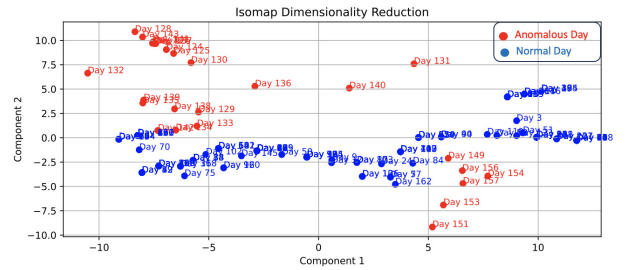


Fig. 5. Isomap Graph for (User 1)

Figure 5 presents a denser plot with more days represented. The points are again separated into two color groups:

Red Points: Mostly concentrated in the upper left section, representing days between Day 128 and Day 140. These points show a noticeable separation from the blue points.

Blue Points: These are more spread across the lower half of the plot, forming a gradient from left to right. The blue points are organized with lower Component 2 values but seem more densely packed toward the right side of the plot, suggesting less variability in their behavior. Red points also appear on the right side, but are fewer in number. Notable red days (Days 153 and 156) appear on the right side, indicating outliers or a potential shift in the data over time.

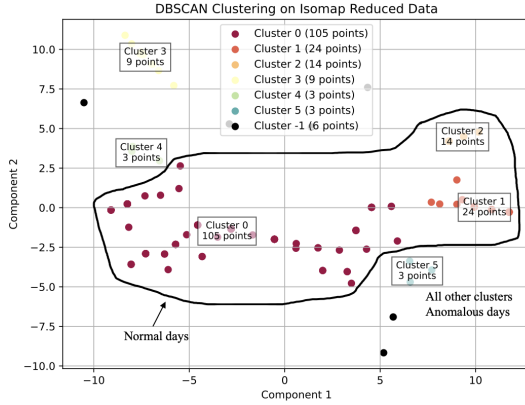


Fig. 6. DBSCAN Graph for (User 1)

Figure 6 illustrates another instance of DBSCAN clustering on the same Isomap-reduced dataset, where the parameter tuning has led to a more detailed division of the data. Here, Cluster 0 comprises 105 points, remaining the most significant group. Other clusters, such as Cluster 1 with 24 points and Cluster 2 with 14 points, represent smaller but still notable groupings. Additionally, Clusters 3 through 5 have between 3 and 9 points each. Notably, the plot also includes Cluster-1, which has 6 points, representing noise points that do not belong to any cluster. This clustering result indicates a more fragmented structure, with distinct clusters and a relatively small number of noise points.

Figure 7 shows SHAP values for day 153, revealing feature contributions to the model's prediction of $f(x) = 1$, deviating from the base value $E[f(x)] = 0.124$, indicating an anomaly. Positive Contributions: USB Disconnect: +0.85 File: +0.18 Logon, Logoff: +0.00 each Negative Contributions: USB Connect: -0.12 HTTP: -0.03 Email: -0.00

The primary contributor to the final prediction is USB Disconnect, indicating the significant importance of this feature. HTTP events result in minor adjustments, both positive and negative, while other features have a negligible impact.

C. Scenario 3: Anomalous User # 375 (GTD0219)

This CERT v4.2 database describes this scenario as follows: "The system administrator becomes disgruntled. Downloads a keylogger and uses a thumb drive to transfer it to his supervisor's machine. The next day, he used the collected

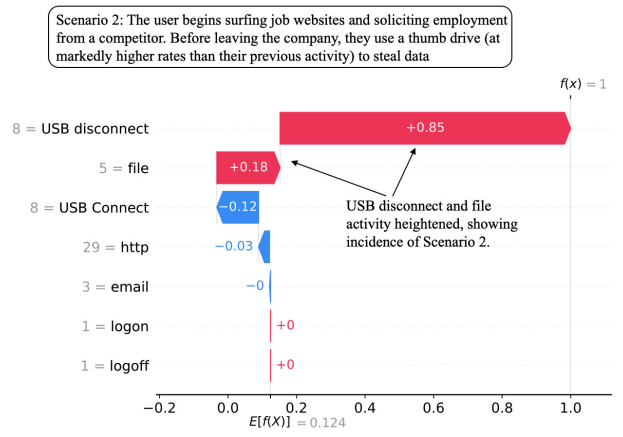


Fig. 7. SHAP Explanation for Scenario 2 (User 1) highlighting feature contributions.

key logs to log in as his supervisor and sent out an alarming mass email, causing panic in the organization. He leaves the organization immediately".

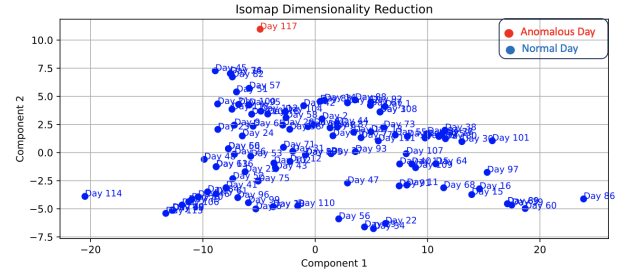


Fig. 8. Isomap Graph for (User 375)

Figure 8, the plot is much denser, with the majority of the points shown in blue, clustered around the middle of the plot. A single red point (Day 117) appears isolated in the upper-left corner, indicating a possible anomaly or a distinct outlier in the dataset.

The blue points are distributed with no distinct separation between clusters, indicating a more uniform distribution in the underlying data. However, there is a notable spread along Component 1, ranging from -20 to 20, with some variability along Component 2 between -7.5 and 10.

Figure 9 presents the DBSCAN clustering result with a different set of parameters applied to the Isomap-reduced dataset. In this case, the algorithm has primarily identified a single, large Cluster 0 comprising 115 points. The presence of Cluster-1 with 3 points indicates that these data points are considered noise or anomalies, as they do not meet the density requirements to form a cluster. The overall structure of this result reveals a single, tightly knit cluster, with minimal noise or outlier points.

Figure 10 shows SHAP values for day 117, revealing feature contributions to the model's prediction of $f(x) = 0.971$, deviating from the base value $E[f(x)] = 0.021$, indicating an anomaly.

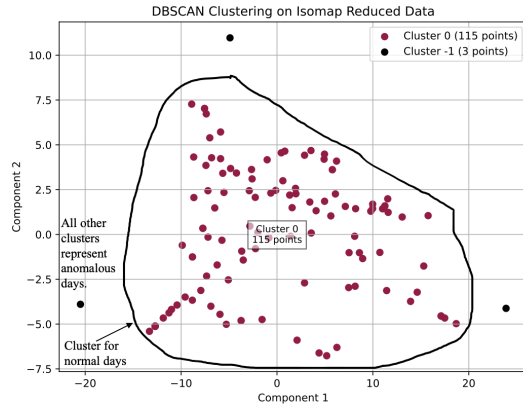


Fig. 9. DBSCAN Graph for (User 375)

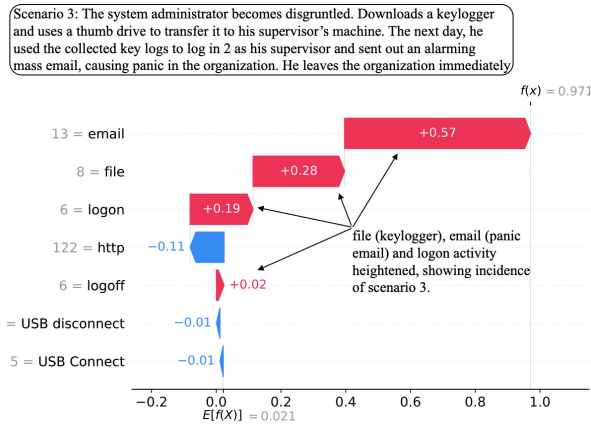


Fig. 10. SHAP Explanation for Scenario 3 (User 375) highlighting feature contributions.

Positive Contributions: Email: +0.57 File: +0.28 Logon: +0.19 Logoff: +0.02

Negative Contributions: HTTP: -0.11 USB Connect: -0.01 USB Disconnect: -0.01

The final prediction score is strongly influenced by email activity, followed by file operations. Logon and logoff events provide moderate positive contributions, while HTTP slightly reduces the score.

VI. DISCUSSION AND CONCLUSION

Our framework, tested on the CERT v4.2 Insider Threat Dataset, uses nonlinear behavioral embeddings and SHAP for explainable anomaly detection. Isomap reduces dimensionality, DBSCAN clusters anomalies (e.g., days 147, 152, 154) [11]. SHAP highlights features like USB activity in data leakage, cutting false positives and aiding GDPR compliance, enhancing transparency.

It visualizes anomalies, delivers insights, and reduces analyst workload. Stable neural network training (near-zero loss by epoch 300) shows generalization. However, SHAP's surrogate model and feature independence assumption may limit sequential data accuracy. Scalability and real-time deployment are challenges.

Future work includes studying SHAP's clarity, adapting to diverse threats, improving SHAP contextualization, and enabling real-time detection. Custom features could boost adaptability.

In conclusion, our approach advances insider threat detection with behavioral embeddings and SHAP, reducing false positives, ensuring compliance, and providing interpretable insights.

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